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10 December 1985
SUPERSEDING
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MILITARY SPECIFICATION

CARTRIDGE, IMPULSE, MARK 45 MOD 0

This specification is approved for use within the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the manufacture and acceptance of the Mark 45 Mod 0 Impulse Cartridge (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

Military

MIL-D-21625	Design and Evaluation of Cartridges for Cartridge Actuated Devices

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-453	Inspection, Radiographic

2.1.2 Drawings and publications. The following drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

Naval Air Systems Command (Code Ident 30003)

DL537926	Cartridge, Impulse, Mark 45 Mod 0

Beneficial comments (recommendations, additions , deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Systems Engineering and Standardization Department (SESD) Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1377

AMSC N/A

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DISTRIBUTION STATEMENT A: Approved for public release; distribution is unlimited.

DL537927 Container, Inner, for Cartridge, Impulse, Mark 45 Type

DL537928 Container, Outer, for Cartridges, Power Sources

PUBLICATIONS

Naval Sea Systems Command (Code Ident 53711)

OP 5 Ammunition and Explosives Ashore

Code of Federal Regulations

49 CFR 100-199 Transportation

(Copies of specifications, standards, other Government documents, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer).

2.2 Order of Precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Conformance to documents. The Mark 45 Mod 0 impulse cartridge covered by this specification shall be manufactured in accordance with DL537926 and all documents listed thereon.

3.3 Materials. All materials used in the manufacture of the cartridges shall conform strictly to the specifications referred to on the respective drawings. When alternate materials or methods of manufacture are specified on the drawings, the bidder's selections shall be clearly stated in the proposal.

3.4 Primary components. For the purpose of this specification the primer mix (Drawing 2406487) is considered as a primary component.

3.5 Insulation resistance. Prior to installation of the bridgewires, the insulation resistance between pins and case shall be not less than 2 megohms at a relative humidity of less than 80 percent when measured with a 500 volt megohmmeter (see 4.6.1).

3.6 Bridgewire. The bridgewire shall be welded to the pins.

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3.7 Bridgewire continuity. Before filling the cavity around the bridges with the igniter compound, the bridge assemblies shall be thoroughly cleaned and inspected for proper attachment and absence of projecting ends. After cleaning, the bridgewire shall be flashed by application of an electric current to assure continuity of both bridgewires. The current shall be regulated to produce a visible red glow of the bridges. An electromagnetic force (EMF) of approximately 1.5 volts ac or dc may be used with a suitable series resistor to adapt the source characteristics to the primer resistance. Flashing is a necessary manufacturing operation and shall be conducted on each bridgewire assembly. Assemblies which fail to glow in both bridges shall be screened from acceptable material in the process. The discarded assemblies may be reworked by replacement of the bridges.

3.8 Primer mix. The primer mix shall be compounded as specified on Drawing 2406487.

3.9 Loading of igniter composition. The techniques of loading the igniter composition shall be such as to avoid damage to the bridgewire. The composition shall be smoothed off flush with the top of the cavity. All igniter composition shall be removed from exterior lateral areas of the spacer. Allow to dry thoroughly before cementing mylar disc in place. The dried composition shall be free from cracks, voids, and separation from the ignition cup. The charge shall entirely cover the bridgewire.

3.10 Propellant charge.

3.10.1 Quantity. All propellant charges shall be determined by weight rather than by volume measurement.

3.10.2 Smokeless powder, handling, and loading. The smokeless powder shall be subjected to no unnecessary handling or exposure to the atmosphere. The handling and loading of smokeless powder exposed to the atmosphere shall be performed only under conditions that do not permit the relative humidity to exceed 75 percent.

3.10.3 Smokeless powder, condition. The smokeless powder used shall exhibit no evidence of decomposition (see 4.6.2)

3.11 Cartridge inspection.

3.11.1 Bridgewire resistance. The resistance of each bridge in each completely assembled cartridge, when measured electrically (see 4.6.4), shall be not less than 0.130 ohm nor greater than 0.160 ohm.

3.11.2 X-ray. Cartridges shall be free of imperfections when X-rayed (see 4.6.5).

3.12 Ballistic requirements. Cartridges shall meet the ballistic requirements specified in 3.12.1 through 3.12.3 when subjected to the test prescribed in 4.6.6.

3.12.1 Maximum pressure. The upper (U) and lower (L) limits of maximum pressure shall be U = 2300 psi and L = 1800 psi, respectively.

3.12.2 Misfire. There shall be no misfires (see 4.6.6.1).

3.12.3 Hangfire. There shall be no hangfires (see 4.6.6.2).

3.13 Workmanship. Cartridges shall be free of the following defects: burrs, dents, incomplete or damaged threads, improper crimp, perforated closure disc, bent pins, or other defects which may prevent entry of the cartridge into the firing chamber, or affect the resistance of the cartridge to moisture. All cartridge marking and identification shall be verified as proper (see 4.6.3).

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4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 **Responsibility for compliance.** All items must meet all requirements of sections 3 and 5. The in spection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 **Classification of inspections.** The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)
b. Quality conformance inspection (see 4.5).

4.3 **Inspection conditions.** Unless otherwise specified (see 6.2), all inspections shall be performed as specified in the applicable test method of 4.6.

4.4 **First article inspection.** First article inspection shall be performed after award of contract and prior to production by a facility designated in the contract or order (see 6.2). A first article sample acceptable for environmental and function testing shall be considered as a sample which has met the requirements of 3.2 through 3.11.2. and 3.13. First article inspection shall be performed on sample units which have been produced using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials, including packaging and packing shall be the same as used for regular production and shall be from the intended sources of supply.

4.4.1 **Sample size.** The first article sample shall consist of 70 cartridges.

4.4.2 **Inspection routine.** Sixty cartridges shall be expended in the tests listed below and 10 shall be retained for investigative purposes. Testing shall consist of the following tests as prescribed MIL-D-21625D: 15 "g" shock (6); vibration (6); temperature and humidity cycling (12). After environmental treatment, these cartridges are to be conditioned together with 12 untreated cartridges, at 70°F and fired as specified in 4.6.5.1. In addition 12 cartridges each shall be conditioned at −65°F and 160°F and fired as specified in 4.6.5.1 to check temperature sensitivity. Acceptance of the 70°F firings shall be in accordance with 4.6.5.1.

4.4.3 **Failure.** Failure of any cartridge shall be cause for rejection of the first article sample.

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4.5 Quality conformance inspection. Quality conformance inspection shall consist of the tests specified in table I.

TABLE I. Quality conformance tests and examinations.

Item	Nature of test	Requirement para.	Test para.	Classification (WR-43)
Body and pin assembly	Insulation resistance	3.5	4.6.1	(M101)
Cartridge	Visual inspection	3.11.3	4.6.3	(M102)
Cartridge	Bridgewire resistance	3.11.1	4.6.4	(C1)
Cartridge	X-ray	3.11.2	4.6.5	(C2)
Cartridge	Ballistics	3.12	4.6.6	(C3)
Cartridge	Preservation-packaging, packing and marking	Section 5	4.6.3	(M103)

4.5.1 Lot size. Unless otherwise specified (see 6.2), the production lot size shall be not less than 111 and not greater than 500. Only primary components from a single lot shall be used in a production lot of cartridges; however, one primary component production lot may be used in more than one cartridge production lot.

4.5.2 Sampling. Unless otherwise specified (see 6.2), sampling shall be in accordance with table II.

TABLE II. Quality conformance sample size.

Lot size	Sample size	Retained sample
66 - 110	15	10
111 - 180	15	10
181 - 300	20	10
301 - 500	25	10
501 - 800	30	10
801 - 1300	35	10
1301 - 3200	40	10
3201 - 8000	50	10
8001 - 22000	75	10

4.5.3 Classification of characteristics. The characteristics verified by the tests and examinations herein are classified as critical, major, or minor in accordance with WR-43. Tests and examinations that verify critical characteristics are identified by the symbol (C) and major characteristics by the symbol (M). The number following the classification symbol indicates the serial number of the test or examination. Tests and examinations which are not annotated with a classification code are classified minor.

4.6 Tests.

4.6.1 Insulation resistance. Prior to assembly of the bridgewire in the body and spacer assembly, the insulation resistance shall be checked with a 500-volt megohmmeter. Units with a resistance of less than 2 megohms shall be cause for rejection.

4.6.2 Smokeless powder, condition. Smokeless powder shall be inspected in accordance with the applicable paragraphs of OP 5. Evidence of decomposition shall be cause for rejection.

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4.6.3 Visual examination. All cartridges shall be visually examined. Cartridges having any of the defects of 3.11.3 shall be cause for rejection. The preservation-packaging, packing, and marking of the cartridges shall be examined for conformance to section 5.

4.6.4 Bridgewire resistance. Conformance with the bridge circuit resistance of 3.11.1 shall be ascertained on all completely assembled cartridges by means of a test circuit which limits the bridge circuit current to 50 milliamperes, maximum. A safety chamber with an interlock switch shall be used to protect the operator during the application of current. A suitable instrument which is known to be accurate to within 0.005 ohm in the range of resistance specified shall be used to determine whether or not the requirement is being met. Units which fail to meet the requirements of 3.11.1 shall be cause for rejection. The results of bridge circuit resistance measurements shall not be affected by resistance of the test circuit and connector.

4.6.5 Radiographic examination. All cartridges, in the first article and production test samples, shall be radiographically examined in accordance with MIL-STD-453. Cartridges having any observable imperfections in assembly shall be cause for rejection. The cartridges shall be positioned on their sides for the most revealing exposure. All cartridges shall be identified with serial numbers prior to examination. The cartridge serial numbers shall be in consecutive order beginning with the number 001 in each production lot. The cartridges shall be arranged on trays or boards in consecutive numerical order, and each radiograph shall carry a permanent identification of the cartridges displayed thereon. The radiograph identification shall include the complete lot number, as stamped on the cartridges, and the span of serial numbers displayed. If any assembly or loading defects are noted in the production lot sample, the contractor shall X-ray the represented lot 100 percent and make available the X-rays to the cartridge design activity for review. Discontinuities in serial numbers shall be identified on the X-ray. Defective cartridges found by the vendor shall be clearly identified on the X-ray and those defectives removed from the production lot. The serial numbers shall be pen or ink stamped on the cartridges with approximately 1/8-inch-high characters. The serial numbers shall be located apart from the other cartridge markings.

4.6.6 Ballistic test. The number of cartridges selected from each production lot for the ballistic test shall be in accordance with table II. The cartridges shall be conditioned for 24 hours at 70°F ± 5°F and fired in a closed cylindrical chamber having the ratio of length to diameter of approximately 1 and an internal volume, exclusive of cartridge, of 10 cm^3 (0.610 in^3), and adapted for the measure of pressure. Firing current shall be not less than 7 amperes to one bridge. Lots which meet the maximum pressure requirements of 3.12.1 at an AQL of .10 shall be acceptable. In addition to the sample of cartridges required for the above test, 10 additional cartridges shall be furnished from the production lot for investigative purposes. Acceptance criteria will be accept on 0 and reject on 1 regardless of lot size.

4.6.6.1 Misfire. The failure of any cartridge to fire shall be cause for rejection of the lot represented, unless such failure is plainly attributable to faulty test equipment or procedure.

4.6.6.2 Hangfire. One or more hangfires (see 6.4.1) shall be cause for rejection of the lot represented.

4.7 Retest. There shall be no retests.

4.7.1 Test failure. If test failure is attributable to an assignable cause, excluding the test cartridge, original test results should be discarded and that portion of the test reconducted.

5. PACKAGING

5.1 Packaging.

5.1.1 Level A. In addition to the following, Level A packaging shall be in accordance with 49 CFR 171-178.

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5.1.1.1 **Inner container.** Four cartridges shall be packaged in accordance with DL537927.

5.2 **Packing.**

5.2.1 **Level A.** In addition to the following, Level A packing shall be in accordance with 49 CFR 171-178.

5.2.1.1 **Outer container.** Twelve inner containers, as described in 5.1.1.1, shall be packed in accordance with DL537928.

5.3 **Marking.** Cartridge marking shall be as specified on Drawing 2240773. In addition, packaging and packing markings shall be in accordance with 49 CFR 171-178 and MIL-STD-129.

6. NOTES

6.1 **Intended use.** The Mark 45 Mod 0 impulse cartridge is intended for use in the fuel valving system of the XXD2B target.

6.2 **Ordering data.** Acquisition documents should specify the following:

a. Title, number and date of this specification
b. Whether first article inspection is required, and if so, specify the test activity
c. Date and place of delivery
d. Inspection conditions, if other than as specified (see 4.3)
e. Lot size, if other than as specified (see 4.5.1)
f. Sampling, if other than as specified (see 4.5.2)
g. Special marking, if any (see 5.3)
h. That the safety precaution requirements of the "Contractor's Safety Manual for Ammunition, Explosives, and Related Dangerous Material," DoD 4145.26M are applicable. NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of "Ammunition and Explosives Ashore," OP5 are applicable.

6.3 **First article.** When a first article inspection is required, the item should be a first article sample. The first article should consist of 70 cartridges. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirements for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

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6.4 Definition.

6.4.1 Hangfire. A hangfire is defined as a firing in which the time from application of current to case opening or first indication of pressure exceeds 0.75 second.

6.5 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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